

3.2 Video Correction

In [Video 3.2 Fitting Lines and Curves](#), from 9:00 to 10:05, $\frac{\partial E}{\partial c} = 0$ should be replaced by $\frac{\partial E}{\partial d} = 0$ (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

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Fitting Curves to Edges

Given: Edge Points (x_i, y_i)

Task: Find polynomial
 $y = f(x) = ax^3 + bx^2 + cx + d$
that best fits the points

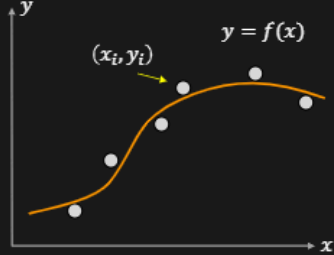
Minimize:

$$E = \frac{1}{N} \sum_i (y_i - ax_i^3 - bx_i^2 - cx_i - d)^2$$

Solve the Linear System Using Least Squares Fit by:

$\frac{\partial E}{\partial a} = 0$
 $\frac{\partial E}{\partial b} = 0$
 $\frac{\partial E}{\partial c} = 0$
 $\frac{\partial E}{\partial d} = 0$

Closed-form solution cumbersome when unknowns are many



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